

This Month's DVD Contents

This month's DVD is a collection of all types of open source engineering tools available freely with one common distribution platform

SNEHA AMBASTHA

CAELinux (version 2013)

CAELinux is a distribution platform with a complete set of engineering tools that are available freely for Linux users. Being built on an updated Xubuntu 12.04 long-term support (LTS) 64-bit base, it focuses on upgrading all its software, developing additional wizards and expanding distribution capabilities. The latest version of this software is basically a mature version of Linux.

Deeds (version 1.80.100.1)

Digital Electronics Education and Design Suite, or Deeds, has been designed to combine logic networks from simple gates to encoders, decoders, multiplexers and demultiplexers. It supports sequential logic networks (like simple flip-flops, registers and counters), finite state machine design and microcomputer programming (at assembly level). The internal file version of Deeds has changed to 1.024, so files saved with the current version (1.80) cannot be opened in any previous versions. However, backward compatibility of files still exists.

PSipce (version 9.1)

This is a simulation program to model the behaviour of a circuit containing analogue devices. This student version of the software can be used as a software based breadboard for testing and refining designs without even touching a hardware component. It comes with a number of built-in models with parameters that one can tweak for a given device.

TARGET 3001

TARGET 3001 is a computer-aided design (CAD)/computer-aided engineering (CAE) tool used for design-

ing printed circuit boards (PCBs). Within a single user interface (UI), it has features like layout, auto-router, schematic, simulation, three-dimensional (3D) view and front-panel design. It allows one to toggle between schematic and layout, or have both in the separate windows beside each other. This is the demo version and can be installed on 32-bit Windows NT/2000/ME/XP/Vista/Win7.

IntegraXor SCADA (version 4.2)

As the name suggests, IntegraXor SCADA is a supervisory control and data acquisition (SCADA) tool that runs on Windows server 2008 and requires about 2.50GB of memory space. It is purely Web based, with HTML5 and CSS3 support. It can be used to remotely view, monitor and control devices, processes and facilities. It can also support a large set of protocols such as Modbus, OPC, Profinet and SQL. It has zero startup cost with free development tools but requires the purchase of runtime licences when ready for deployment.

V-USB (version 1.1)

V-USB is the software implementation of a low-speed USB device for Atmel's AVR microcontrollers (MCUs). It makes USB hardware building possible with almost any AVR MCU, without any additional chip. This freeware supports multiple endpoints to some degree, where usually USB specification forbids bulk endpoints for low-speed devices. It runs on any AVR MCU with at least 2kB of flash memory, 128 bytes random access memory (RAM) and a clock rate of at least 12MHz.

Popular resources

- 7Zip. Version 9.38 (file archiver)
- Java Runtime Environment. Version 8.0.450.15 (runs programs written in Java programming language)
- VLC. Version 2.2.1 (media player)
- Opera. Version 29.0.1795.47 (Web browser)
- Free Download Manager. Version 3.9.5 (file download manager)
- Thunderbird. Version 31.1.0 (email application)
- Mozilla Firefox. Version 37.0.2 (Web browser)
- Apache Open Office. Version 4.1.1 (open office)

Digital logic simulator (version 0.4)

This freeware simulates computer chips. It allows users to build computer components like RAM, Adders and ALU. It can help them build a scene, save it and import it as a single chip. Usually, with complexity of scenes, the number of resources to be used increases and the software becomes slow. However, the chip-importing feature in this simulator increases its efficiency. It is compatible with any version of Windows.

Essentia (version 2.0)

Essentia is a C++ library for audio analysis and audio based music-information retrieval. This open source software is released under Affero GPL licence and contains a large collection of reusable algorithms that implement audio input/output functionality, standard digital signal processing blocks, statistical characterisation of data, and a large set of spectral, temporal, tonal and high-level music descriptors. The library is cross-platform and supports almost all operating systems. It is wrapped in Python and includes a number of predefined executable extractors for available music descriptors. ●

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